Yiyang CAI

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Hong Kong University of Science and Technology - Clear Water Bay - Hong Kong

Education

- **Hong Kong University of Science and Technology** *Sep. 2024 - Current*
  Ph.D. Student of Independent Interdisciplinary Program (Supervisor: Prof. Wenhan LUO and Prof. Yike GUO) Kowloon, HKSAR
  - Research Interests: Image/Video Generation, Multimodal Learning, World Models

- **University of California, Berkeley** *Aug. 2020 - Dec. 2021*
  Master of Engineering, Visual Computing and Computer Graphics, (Supervisor: Prof. Brian BARSKY) Berkeley, CA, United States
  - GPA: 3.65/4.00

- **Beihang University** *Aug. 2016 - Jun. 2020*
  Bachelor of Engineering, Automation (Shenyuan Honor School) Beijing, China
  - GPA: 3.81/4.00 (Outstanding Graduate Honor)

Publications (First or Co-first are lighted in red)

[1] **Y. Cai***, N. Chen*, R. Xie, J. Pan, C. Jiang, C. Chen. C. Liu, Z. Sun, W. Xue, W. Luo & Y. Guo. HOMIE: Human-object Centric Video Personalization via Multimodal Intelligent Enhancement. The 40th Annual Conference on Neural Information Processing Systems, 2026. (CCF-A)

[2] **Y. Cai***, N. Chen*, et al. StreamingR2V: Towards Long and Interactive Reference-to-Video Generation. Under Review, 2026.

[3] N. Chen*, **Y. Cai***, R. Xie, J. Pan, C. Chen. W. Jia, Z. Cheng, C. Liu, Z. Sun & W. Luo. DomainShuttle: Freeform Open Domain Subject-driven Text-to-video Generation. Under Review, 2026.

[4] C. Jiang, Y. Zhang, **Y. Cai**, C. Chan, Y. Liu, M. Chen. W. Xue & Y. Guo. Semantic Voting: A Self-Evaluation-Free Approach for Efficient LLM Self-Improvement on Unverifiable Open-ended Tasks. The 14th International Conference on Learning Representations, 2026. (CCF-A)

[5] Y. Liu, H. Xu, **Y. Cai**, C. Jiang, S. Han & Y. Guo. Modeling the Brain's Grammar: ROI-Guided fMRI Pretraining for Transferable and Interpretable Vision Decoding. IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026. (CCF-A)

[6] **Y. Cai**, Z. Jiang, Y. Liu, C. Jiang, W. Xue, Y. Guo & W. Luo . Foundation Cures Personalization: Improving Personalized Models' Prompt Consistency via Hidden Foundation Knowledge. The 39th Annual Conference on Neural Information Processing Systems, 2025. (CCF-A)

[7] C. Jiang, C. Chan, **Y. Cai**, Y. Liu, W. Xue, & Y. Guo. Graceful Forgetting in Generative Language Models. Empirical Methods in Natural Language Processing (Main Conference), 2025. (CCF-B)

[8] W. Cheng, W. Zhang, H. Shen, **Y. Cai**, X. He, K. Lv, & Y. Liu . Optimize weight rounding via signed gradient descent for the quantization of LLMs. Empirical Methods in Natural Language Processing, 2024.

[9] H. Chang, H. Shen, **Y. Cai**, X. Ye, Z. Xu, W. Cheng, K. Lv, W. Zhang, Y. Lu & H. Guo. Effective quantization for diffusion models on CPUs. NeurIPS 2023 Workshop on Diffusion Models, 2023.

[10] L. Tang*, **Y. Cai***, J. Liu, Z. Hong, M. Gong, M. Fan, J. Han, J. Liu, E. Ding, & J. Wang. Few-shot font generation by learning fine-grained local styles. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 7895-7904), 2022. (CCF-A)

Projects

- **Multi-subject-driven Video Generation** *Aug. 2025 - Sep. 2026*
  Keyword: Personalized Diffusion Models, Video Generation, Human-object Interaction [GitHub]
  - Completed a comprehensive literature review of video generation approaches.
  - Inspired by unified understanding and generation models, proposed a MLLM integration strategy to enhance video personalization models in more challenging scenarios, including multi-subject video personalization, abstract concept personalization.
  - Conducted comprehensive comparison to popular video personalization models, validated the advantages of the proposed method and the corresponding paper submission is now accepted by NeurIPS 2026 as a poster.
  - Developed an autoregressive framework that extends short-horizon video personalization models to support long-horizon and interactive video personalization. Designed an attention-enhancement strategy to decouple identity information between historical context and reference conditions. Completed experimental validation and submitted the work for publication.

- **Training-free Prompt Following Optimization for Facial Personalization Models** *Sep. 2024 - Apr. 2025*
  Keyword: Personalized Diffusion Models, Prompt Following, Training-free [ ]
  - Completed a comprehensive literature review of facial personalization approaches.
  - Identified the current limitation shared by state-of-the-art facial personalization methodologies: poor prompt following performance on fine-grained facial attributes (e.g., hair style, accessories, expression).
  - Proposed a dual-denoising branch in personalization models (w./w.o. identity information) with an optimized self-attention modules that transfers satisfying facial attributes into personalization denoising process.
  - Validated the proposed method on popular personalization models built upon different foundation models (e.g., SDv1.5, SDXL, FLUX.1-dev).
  - Finished a paper submission and the paper has been accepted by NeurIPS 2026.

Working Experience

- **ByteDance Inc. (China)** [ ] *Aug. 2025- Sep. 2026*
  Algorithm Intern Beijing, China
  - Explore video personalization under the enhancement of MLLMs.
  - Explore infinite video generation under auto-regressive architecture and its application in video personalization, including interactive reference insertion.
- **Intel Inc. (China)** [ ] *Jun. 2022- Jun. 2024*
  Machine Learning Engineer, Group of Data Center and Artificial Intelligence Shanghai, China
  - Developed and maintained key features of **Intel Neural Compressor**, including quantization and pruning for diffusion models and large language models.
  - Developed efficient GPTQ API for a large variety of LLMs and VLMs (e.g., MPTs, Qwens, ChatGLMs, Baichuan, LLaVA). Designed pruning API to support different orthogonal configurations (structures, criterion, schedulers, etc.). Validated API's function on a wide range of model, including CV, NLP models and popular LLMs.
- **Baidu Inc.** [ ] *Sep. 2020 - Nov. 2021*
  Algorithm Intern, VIS Group Beijing, China
  - GAN-based style transfer research for few-shot Chinese font generation: given a small set of style-specified glyphs ($\leq$ 100 examples), generated the entire personalized font ($\approx$ 20K instances). Applied a novel attention machanism to improve the model's ability of capturing fine-grained style and spatial feature representations from glyphs. Designed character decomposition strategy to boost efficiency of learning characters' component representations.

Skills

- **Programming Languages:** Python, C++
- **Deep Learning & Computer Vision Tools:** PyTorch, DeepSpeed, Matplotlib, OpenCV and Pandas
- **Language Proficiency:** TOEFL: 104 (Reading 28 + Listening 25 + Speaking 26 + Writing 25)

Teaching Experience

- **Teaching Assistant** *Feb. 2026 - May. 2026*
  TA duty for EMIA-4110 (Practical Machine Learning) (Teacher: Prof. Wenhan LUO) HKUST
  - Delivered several presentations of advanced artificial intelligence.
  - Supported the professor's course preparation of image generation tutorials.
- **Teaching Assistant** *Sep. 2025 - Dec. 2025*
  TA duty for MAIE-5421 (Computer Vision) (Teacher: Prof. Wenhan LUO) HKUST
  - Delivered several presentations of advanced artificial intelligence.
  - Supported the professor's course preparation of tutorials, project designing and final grading.
- **Teaching Assistant** *Feb. 2025 - May. 2025*
  TA duty for EMIA-4110 (Practical Machine Learning) (Teacher: Prof. Wenhan LUO) HKUST
  - Delivered several presentations of using Colab and PyTorch to undergraduate students.
  - Supported the professor's course preparation of image generation tutorials.

Service

Reviewer of CVPR 2026, ECCV 2026, NeurIPS 2026, AAAI 2027, ICLR 2027.